

Roll No: _____

Name: _____

Instructions!

- This quiz is an individual activity, you are strictly NOT allowed to talk with your fellow colleagues, even not allowed to share resources i.e. pen, calculator and/or rough sheets.
- Understanding the questions is, part of the quiz. So, none of your queries will be entertained during quiz time.
- Solve all the questions on the provided space in your question paper.
- This quiz will be collected back after 30 minutes.
- Anyone caught in an act of dishonesty would be awarded an "F" grade in this quiz.

Question 01:

[05 Marks]

Choose the best answer:

1. Which statement is TRUE about Stochastic Gradient Descent?
A) Uses one training example per update
B) Uses full dataset for each update
C) Does not guarantee convergence to the global minima
D) Requires no cost function
2. In multivariate linear regression, the hypothesis in vectorized form is:
A) $h\theta(x) = \theta_0 + \theta_1 x_1$
B) $h\theta(x) = \theta^T x$
C) $h\theta(x) = \theta^T x^T$
D) $h\theta(x) = \theta_0 + \theta^T x$
3. If the learning rate is too large, gradient descent may:
A) Converge very slowly
B) Immediately reach global minimum
C) Diverge or fluctuate
D) Reduce feature importance
4. Which is an advantage of the Normal Equation over Gradient Descent?
A) Faster for extremely large datasets
B) Requires feature scaling
C) No need to choose learning rate
D) Needs many iterations
5. In supervised learning, the training data is:
A) Unlabeled
B) Randomly generated
C) Labeled with correct outputs
D) Noise-free

Question 02:

[05 Marks]

1. What is Feature Engineering? Explain its purpose and discuss atleast three commonly used feature engineering techniques. [5]

2. Suppose you're training a model and observe this perttren: [5]

No. of iterations	Cost/loss
100	150
200	170
300	210

Explain what is happening and what is the likely cause of this observation? Also, suggest two solutions to this problem.

3. You are training a model with two features, where, [4]

- Feature 1: Salary (20,000 – 2,000,000)
- Feature 2: Experience (1 – 10)

What issue may occure? Explain geometrically how this affects gradient descent.

4. Given hypothesis:

$$h\theta(x) = \theta_0 + \theta_1 x_1$$

Where:

$$\theta_0 = 10, \theta_1 = 35$$

For $x_1 = 4$

(a) Compute predicted value.

[2]

(b) If actual value is 20, compute squared error.

[2]

5. You are given the following dataset:

[5]

x	y
1	3
2	5
3	7
4	9
5	11

Using the Normal Equation, compute the optimal values of θ_0 and θ_1 .